Homework 3

High-Dimensional Probability (HDP)

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Problems

Problem 1 (1)

Let $f : \Theta \times \mathcal{X} \to \mathbb{R}$ be such that for every $x \in \mathcal{X}$,

$$|f_\theta(x) - f_{\theta'}(x)| \leq C\, d(\theta, \theta'),$$

where d is a metric on Θ. Define the function class

$$\mathcal{F} := \{f_\theta : \theta \in \Theta\}.$$

Show that

$$\mathcal{N}(\mathcal{F}, \|\cdot\|_\infty, \varepsilon) \leq \mathcal{N}\Big(\Theta, d, \frac{\varepsilon}{C}\Big).$$

Solution. Define the parametrization map

$$T : \Theta \to (\mathbb{R}^{\mathcal{X}}, \|\cdot\|_\infty), \qquad T(\theta) = f_\theta.$$

The assumption implies that T is C-Lipschitz from (Θ, d) into the sup-norm space: indeed, for any $\theta, \theta' \in \Theta$,

$$\|T(\theta) - T(\theta')\|_\infty = \sup_{x \in \mathcal{X}} |f_\theta(x) - f_{\theta'}(x)| \leq \sup_{x \in \mathcal{X}} C\, d(\theta, \theta') = C\, d(\theta, \theta').$$

Now take a (ε/C)-net of Θ in the metric d. That is, let $\{\theta_1, \ldots, \theta_N\} \subseteq \Theta$ satisfy

$$\Theta \subseteq \bigcup_{i=1}^{N} B_d\Big(\theta_i, \frac{\varepsilon}{C}\Big), \qquad N = \mathcal{N}\Big(\Theta, d, \frac{\varepsilon}{C}\Big).$$

Consider its image under T:

$$\mathcal{G} := \{f_{\theta_1}, \ldots, f_{\theta_N}\} \subseteq \mathcal{F}.$$

We claim $\mathcal{G}$ is an ε-net for $\mathcal{F}$ in $\|\cdot\|_\infty$. Indeed, for any $f_\theta \in \mathcal{F}$, choose i such that $d(\theta, \theta_i) \leq \varepsilon/C$. Then by the Lipschitz bound,

$$\|f_\theta - f_{\theta_i}\|_\infty = \|T(\theta) - T(\theta_i)\|_\infty \leq C\, d(\theta, \theta_i) \leq C \cdot \frac{\varepsilon}{C} = \varepsilon.$$

So every f_θ lies within sup-norm distance ε of some element of $\mathcal{G}$, hence $\mathcal{G}$ is an ε-cover of $\mathcal{F}$. Therefore,

$$\mathcal{N}(\mathcal{F}, \|\cdot\|_\infty, \varepsilon) \leq |\mathcal{G}| = N = \mathcal{N}\Big(\Theta, d, \frac{\varepsilon}{C}\Big),$$

which is exactly the desired inequality.

Problem 2 (1)

Let $\{X_t\}_{t\in T}$ be a centered sub-Gaussian process indexed by a metric space (T,d), and assume it is (random-)Lipschitz with constant C in the sense that

$$|X_t - X_s| \le C\, d(t,s) \qquad \forall s,t \in T,$$

with $\mathbb{E}[C] < \infty$. Assume also the sub-Gaussian increment condition (one standard form) that for all $s,t \in T$ and all $\lambda \ge 0$,

$$\mathbb{E}\exp\big(\lambda(X_t - X_s)\big) \le \exp\Big(\frac{\lambda^2 d(t,s)^2}{2}\Big).$$

Show that

$$\mathbb{E}\Big[\sup_{t\in T} X_t\Big] \le \inf_{\delta>0}\Big\{2\delta\ \mathbb{E}[C]\ +\ 12\int_\delta^\infty \sqrt{\log \mathcal{N}(T,d,\varepsilon)}\, d\varepsilon\Big\}.$$

Lemma 0.1 (Finite supremum bound for sub-Gaussian variables)**.** *Let $\{Y_i\}_{i=1}^m$ be a finite family of random variables such that*

$$\mathbb{E}\, e^{\lambda Y_i} \le \exp\Big(\frac{\lambda^2\sigma^2}{2}\Big) \qquad \forall \lambda \ge 0,\ \forall i.$$

Then

$$\mathbb{E}\Big[\max_{1\le i\le m} Y_i\Big] \le \sigma\sqrt{2\log m}.$$

Proof sketch. Fix $\lambda > 0$ and set $M = \max_{1\le i\le m} Y_i$. Use the identity

$$M = \frac{1}{\lambda}\log(e^{\lambda M}) = \frac{1}{\lambda}\log\Big(\max_{1\le i\le m} e^{\lambda Y_i}\Big).$$

Then apply Jensen to log and bound $\max_i(\cdot) \le \sum_i(\cdot)$ to get

$$\mathbb{E}\, M \le \frac{1}{\lambda}\log\Big(\sum_{i=1}^m \mathbb{E}\, e^{\lambda Y_i}\Big) \le \frac{1}{\lambda}\log\Big(m\, e^{\lambda^2\sigma^2/2}\Big) = \frac{\log m}{\lambda} + \frac{\lambda\sigma^2}{2}.$$

Now minimize the RHS over $\lambda > 0$ (complete this optimization carefully) to obtain $\mathbb{E}\, M \le \sigma\sqrt{2\log m}$. □

Solution. We build a multiscale approximation of T by nets and telescope along the resulting chain.

1) Nets and projections. Fix a decreasing sequence of radii $\varepsilon_0 > \varepsilon_1 > \cdots > \varepsilon_n > 0$. For each k, let $T_k \subseteq T$ be an ε_k-net, so that

$$|T_k| = \mathcal{N}(T,d,\varepsilon_k).$$

For each $t \in T$, choose a (measurable) projection $\pi_k(t) \in T_k$ satisfying

$$d\big(t,\pi_k(t)\big) \le \varepsilon_k.$$

2) Telescoping decomposition. For any fixed $t \in T$,

$$X_t = X_{\pi_0(t)} + \sum_{k=1}^{n} \big(X_{\pi_k(t)} - X_{\pi_{k-1}(t)}\big) + \big(X_t - X_{\pi_n(t)}\big).$$

Taking $\sup_{t\in T}$ and then expectation gives

$$\mathbb{E}\sup_{t\in T} X_t \le \mathbb{E}\sup_{t\in T} X_{\pi_0(t)} + \sum_{k=1}^{n} \mathbb{E}\sup_{t\in T}\big(X_{\pi_k(t)} - X_{\pi_{k-1}(t)}\big) + \mathbb{E}\sup_{t\in T}\big(X_t - X_{\pi_n(t)}\big).$$

3) The coarse-scale term. Take $\varepsilon_0 \ge \operatorname{diam}(T) := \sup_{s,t\in T} d(s,t)$ so that one ball covers T. Then we may choose $T_0 = \{t_0\}$ and hence $\pi_0(t) = t_0$ for all t. If the process is centered, $\mathbb{E}\, X_{t_0} = 0$, so this term vanishes:

$$\mathbb{E}\sup_{t\in T} X_{\pi_0(t)} = \mathbb{E}\, X_{t_0} = 0.$$

(If your course uses a different centering convention, adjust this step accordingly.)

4) Increment terms via a finite maximum bound. Fix $k \in \{1,\dots,n\}$. For each t the increment is $X_{\pi_k(t)} - X_{\pi_{k-1}(t)}$ where $\pi_k(t) \in T_k$ and $\pi_{k-1}(t) \in T_{k-1}$. So

$$\sup_{t\in T}\big(X_{\pi_k(t)} - X_{\pi_{k-1}(t)}\big) \le \max_{u\in T_k,\ v\in T_{k-1}} (X_u - X_v).$$

Next use the increment sub-Gaussian assumption and the triangle inequality:

$$d(u,v) \le d(u,t) + d(t,v) \le \varepsilon_k + \varepsilon_{k-1},$$

whenever $u = \pi_k(t)$ and $v = \pi_{k-1}(t)$. Hence each random variable $(X_u - X_v)$ appearing above is sub-Gaussian with proxy $\sigma_k := \varepsilon_k + \varepsilon_{k-1}$.

Apply the lemma with index size $|T_k||T_{k-1}|$:

$$\mathbb{E}\sup_{t\in T}\big(X_{\pi_k(t)} - X_{\pi_{k-1}(t)}\big) \le (\varepsilon_k + \varepsilon_{k-1})\sqrt{2\log(|T_k||T_{k-1}|)}.$$

Now use

$$\log(|T_k||T_{k-1}|) = \log|T_k| + \log|T_{k-1}| \le 2\log|T_k| \quad \text{(since the nets get larger as } \varepsilon \text{ decreases)},$$

to get a cleaner bound of the form

$$\mathbb{E}\sup_{t\in T}\big(X_{\pi_k(t)} - X_{\pi_{k-1}(t)}\big) \ \le\ c_1(\varepsilon_k + \varepsilon_{k-1})\sqrt{\log \mathcal{N}(T,d,\varepsilon_k)},$$

where you should compute the exact constant c_1 carefully.

5) Remainder term via Lipschitz. By Lipschitzness,

$$X_t - X_{\pi_n(t)} \le |X_t - X_{\pi_n(t)}| \le C\, d(t,\pi_n(t)) \le C\,\varepsilon_n.$$

Therefore,

$$\mathbb{E}\sup_{t\in T}(X_t - X_{\pi_n(t)}) \le \mathbb{E}[C]\,\varepsilon_n.$$

6) Dyadic choice and entropy integral. Fix $\delta > 0$ and choose a dyadic scale such that

$$\varepsilon_n = 2\delta, \qquad \varepsilon_{k-1} = 2\varepsilon_k \ (k \geq 1),$$

and ε_0 large enough to dominate $\operatorname{diam}(T)$. With this choice, $(\varepsilon_k + \varepsilon_{k-1})$ is comparable to ε_k, and the discrete sum becomes a dyadic Riemann-type sum.

A standard monotonicity argument (since $\varepsilon \mapsto \mathcal{N}(T, d, \varepsilon)$ is nonincreasing) yields an inequality of the form

$$\sum_{k=1}^{n} \varepsilon_k \sqrt{\log \mathcal{N}(T, d, \varepsilon_k)} \ \leq \ c_2 \int_{\delta}^{\infty} \sqrt{\log \mathcal{N}(T, d, \varepsilon)}\, d\varepsilon,$$

for an explicit absolute constant c_2. Track constants through Steps 4–6 to match the target coefficient 12.

7) Combine and optimize over δ. Using $\varepsilon_n = 2\delta$, the remainder bound becomes $2\delta\ \mathbb{E}[C]$. Putting everything together gives

$$\mathbb{E} \sup_{t \in T} X_t \leq 2\delta\ \mathbb{E}[C] + 12 \int_{\delta}^{\infty} \sqrt{\log \mathcal{N}(T, d, \varepsilon)}\, d\varepsilon,$$

and taking $\inf_{\delta > 0}$ finishes.

Problem 3 (1)

Let $X_1, \dots, X_n$ be i.i.d. random vectors in $[0,1]^d$ with law μ, and let $\mu_n := \frac{1}{n}\sum_{i=1}^n \delta_{X_i}$. Define

$$W_1(\mu_n, \mu) := \sup\Big\{ \mathbb{E}_\mu[f(X)] - \mathbb{E}_{\mu_n}[f(X)] \;:\; f : [0,1]^d \to \mathbb{R},\ f \text{ is 1-Lipschitz},\ f(0) = 0 \Big\}.$$

Let $\mathcal{F}_0$ be the class of all 1-Lipschitz $f : [0,1]^d \to \mathbb{R}$ with $f(0) = 0$.

(1) Show $\mathcal{N}(\mathcal{F}_0, \|\cdot\|_\infty, \varepsilon) \le \exp\big(\frac{c}{\varepsilon^d}\big)$ for $0 < \varepsilon < 1$ and $\mathcal{N}(\mathcal{F}_0, \|\cdot\|_\infty, \varepsilon) = 1$ for $\varepsilon \ge 1$ (with $c = c(d)$).

(2) Show the Dudley entropy integral bound diverges when $d \ge 2$.

(3) Show

$$\mathbb{E}[W_1(\mu_n, \mu)] \lesssim \begin{cases} n^{-1/2}, & d = 1, \\ n^{-1/2}\log n, & d = 2, \\ n^{-1/d}, & d \ge 3. \end{cases}$$

(4) If μ has a density on $[0,1]^d$, show for every fixed $x \in [0,1]^d$,

$$\mathbb{E}\Big[\min_{1\le i\le n} \|x - X_i\|\Big] \gtrsim n^{-1/d}.$$

(5) Conclude that if μ has a density, then $\mathbb{E}[W_1(\mu_n, \mu)] \gtrsim n^{-1/d}$.

Lemma 0.2 (External nets dominate internal nets)**.** *Let (K, d) be a metric space. For every $\varepsilon > 0$,*

$$\mathcal{N}(K, d, \varepsilon) \le \mathcal{N}_{\text{ext}}\Big(K, d, \frac{\varepsilon}{2}\Big),$$

where $\mathcal{N}_{\text{ext}}$ allows centers outside K.

Proof. Let $\{p_1, \dots, p_m\}$ be an external $(\varepsilon/2)$-net of K with minimal size. For each i, the ball $B(p_i, \varepsilon/2)$ must intersect K (otherwise p_i is removable). Pick $q_i \in K \cap B(p_i, \varepsilon/2)$ and set $Q = \{q_1, \dots, q_m\} \subseteq K$. For any $x \in K$, choose i with $d(x, p_i) \le \varepsilon/2$, then

$$d(x, q_i) \le d(x, p_i) + d(p_i, q_i) \le \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

So Q is an internal ε-net and $|Q| = m$, giving the claim. □

Solution. (1) Covering $\mathcal{F}_0$ in sup norm (grid + discrete increments). The high-level move is: discretize the cube by a grid of mesh comparable to ε, then discretize *allowed changes* between neighboring grid points in steps of size ε.

Step A: trivial large-ε case. Since f is 1-Lipschitz and anchored, $|f(x)| \le \operatorname{diam}([0,1]^d)$. Hence for ε larger than a dimension-dependent constant, one function (e.g. 0) covers. (Your problem statement normalizes this threshold to $\varepsilon \ge 1$.)

Step B: build a grid. Let $\eta := \varepsilon/(4\sqrt{d})$ and define the lattice

$$G := \{0, \eta, 2\eta, \dots, \lfloor 1/\eta \rfloor \eta\}^d \subset [0,1]^d, \qquad |G| \le \Big(1 + \frac{1}{\eta}\Big)^d \le \Big(\frac{c_0}{\varepsilon}\Big)^d,$$

for a constant $c_0 = c_0(d)$.

Step C: define a finite approximating class. Consider functions g that are: (i) constant on each grid cell, (ii) satisfy $g(0) = 0$, (iii) take values in $\varepsilon\mathbb{Z}$ on grid vertices, and (iv) obey a discrete Lipschitz constraint along grid edges: for neighboring vertices $u \sim v$,

$$|g(u) - g(v)| \le \varepsilon.$$

Call this finite family $\mathcal{G}$.

Counting: View the grid as a connected graph on $|G|$ vertices. Starting from the fixed value at the origin, each time you expose a new vertex across an edge, the increment has at most 3 admissible choices (down/same/up by ε). This crude exploration bound yields

$$|\mathcal{G}| \le 3^{|G|-1} \quad \Rightarrow \quad \log|\mathcal{G}| \le (\log 3)\,|G| \le \frac{c_1}{\varepsilon^d}.$$

Therefore $|\mathcal{G}| \le \exp(c_1/\varepsilon^d)$.

Approximation: Given $f \in \mathcal{F}_0$, define g on grid vertices by rounding f to $\varepsilon\mathbb{Z}$ (and then extend cellwise-constant). You need to check two things:

- the rounded values still satisfy the neighbor constraint (use Lipschitzness + choice of η);
- for any x, letting u be the cell vertex for x, you have $|f(x)-g(x)| \le |f(x)-f(u)|+|f(u)-g(u)| \le \varepsilon/2 + \varepsilon/2 = \varepsilon$.

Conclude $\mathcal{G}$ is an ε-net in $\|\cdot\|_\infty$, so

$$\mathcal{N}(\mathcal{F}_0, \|\cdot\|_\infty, \varepsilon) \le |\mathcal{G}| \le \exp\Big(\frac{c}{\varepsilon^d}\Big), \qquad 0 < \varepsilon < 1.$$

(2) Dudley integral divergence for $d \ge 2$. If $\log\mathcal{N}(\mathcal{F}_0, \|\cdot\|_\infty, \varepsilon) \lesssim \varepsilon^{-d}$, then

$$\int_0^1 \sqrt{\log\mathcal{N}(\mathcal{F}_0, \|\cdot\|_\infty, \varepsilon)}\, d\varepsilon \;\gtrsim\; \int_0^1 \varepsilon^{-d/2}\, d\varepsilon,$$

which diverges exactly when $d/2 \ge 1$, i.e. $d \ge 2$.

(3) Upper bounds on $\mathbb{E}\,W_1(\mu_n, \mu)$. Define the centered empirical process indexed by $f \in \mathcal{F}_0$:

$$Z_f := \frac{1}{n}\sum_{i=1}^n \Big(f(X_i) - \mathbb{E}\,f(X)\Big).$$

Then by the dual definition,

$$W_1(\mu_n, \mu) = \sup_{f\in\mathcal{F}_0} \big(\mathbb{E}\,f(X) - \tfrac{1}{n}\sum_{i=1}^n f(X_i)\big) = \sup_{f\in\mathcal{F}_0}(-Z_f),$$

so bounding $\mathbb{E}\sup_{f\in\mathcal{F}_0} Z_f$ controls $\mathbb{E}\,W_1$ up to constants.

Sub-Gaussian increments: For $f, g \in \mathcal{F}_0$, each summand in $Z_f - Z_g$ is bounded by $2\|f-g\|_\infty/n$, so Hoeffding's lemma gives (for all $\lambda \ge 0$)

$$\mathbb{E}\exp\big(\lambda(Z_f - Z_g)\big) \le \exp\Big(\frac{\lambda^2}{2}\cdot\frac{\|f-g\|_\infty^2}{n}\Big).$$

Thus $\{Z_f\}$ is sub-Gaussian with metric

$$d_n(f,g) := \frac{\|f-g\|_\infty}{\sqrt{n}}.$$

Also it is Lipschitz in f with deterministic constant $C = 2$:

$$|Z_f - Z_g| \le \frac{1}{n}\sum_{i=1}^{n} |f(X_i) - g(X_i)| + \mathbb{E}\,|f(X) - g(X)| \le 2\|f-g\|_\infty.$$

Entropy under d_n:

$$\mathcal{N}(\mathcal{F}_0, d_n, \varepsilon) = \mathcal{N}(\mathcal{F}_0, \|\cdot\|_\infty, \sqrt{n}\,\varepsilon), \quad \log \mathcal{N}(\mathcal{F}_0, d_n, \varepsilon) \lesssim \frac{1}{(\sqrt{n}\,\varepsilon)^d} = n^{-d/2}\varepsilon^{-d}.$$

Moreover $\log \mathcal{N} = 0$ once $\sqrt{n}\,\varepsilon \gtrsim 1$ (i.e. $\varepsilon \gtrsim n^{-1/2}$).

Case $d = 1$: Dudley's integral converges:

$$\mathbb{E}\sup_f Z_f \lesssim \int_0^{n^{-1/2}} \sqrt{n^{-1/2}\varepsilon^{-1}}\, d\varepsilon \asymp n^{-1/2}.$$

Case $d = 2$: Use the *improved chaining* bound from your previous problem:

$$\mathbb{E}\sup_f Z_f \le \inf_{\delta>0}\left\{2\delta\ \mathbb{E}[C] + 12\int_\delta^\infty \sqrt{\log \mathcal{N}(\mathcal{F}_0, d_n, \varepsilon)}\, d\varepsilon\right\}.$$

Here $\mathbb{E}[C] = 2$ and the integral truncates at $n^{-1/2}$, giving

$$\int_\delta^{n^{-1/2}} \sqrt{n^{-1}\varepsilon^{-2}}\, d\varepsilon \asymp n^{-1/2}\int_\delta^{n^{-1/2}} \frac{d\varepsilon}{\varepsilon} = n^{-1/2}\log\Big(\frac{n^{-1/2}}{\delta}\Big).$$

Pick (for example) $\delta = n^{-1}$ to get the stated $n^{-1/2}\log n$ rate.

Case $d \ge 3$: In the improved chaining bound, choose $\delta = n^{-1/d}$. Then $\delta \ge n^{-1/2}$, so the entropy integral term is 0 (net size = 1 at those scales), and the bound becomes $\mathbb{E}\sup_f Z_f \lesssim \delta \asymp n^{-1/d}$.

Finally, convert $\mathbb{E}\sup_f Z_f$ to $\mathbb{E}\,W_1(\mu_n, \mu)$ (symmetry of $\pm Z_f$).

(4) Lower bound on the nearest-neighbor distance (density case). Fix $x \in [0,1]^d$ and set $R := \min_{1\le i\le n}\|x - X_i\|$. Use the layer-cake identity:

$$\mathbb{E}[R] = \int_0^\infty \mathbb{P}(R > r)\, dr.$$

Now

$$\mathbb{P}(R > r) = \mathbb{P}\left(X_i \notin B(x,r)\ \forall i\right) = \big(1 - \mu(B(x,r))\big)^n.$$

If μ has density p bounded by $\|p\|_\infty \le L$, then

$$\mu(B(x,r)) \le L\,\mathrm{vol}(B(0,1))\, r^d = L v_d r^d.$$

Choose $r_0 := (2Lv_d n)^{-1/d}$ so that $Lv_d r_0^d = \frac{1}{2n}$. Then for $0 \le r \le r_0$,

$$\mathbb{P}(R > r) \ge \left(1 - \frac{1}{2n}\right)^n \ge e^{-1}.$$

Hence

$$\mathbb{E}[R] \ge \int_0^{r_0} e^{-1}\, dr = e^{-1} r_0 \asymp n^{-1/d}.$$

(5) Lower bound on $\mathbb{E}\, W_1$ (density case). Let $S_n = \{X_1, \dots, X_n\}$ and define

$$h_n(x) := \operatorname{dist}(x, S_n) - \operatorname{dist}(0, S_n), \qquad \operatorname{dist}(x, S_n) := \min_{1 \le i \le n} \|x - X_i\|.$$

Then h_n is 1-Lipschitz and $h_n(0) = 0$, so $h_n \in \mathcal{F}_0$. Therefore

$$W_1(\mu_n, \mu) \ge \mathbb{E}_\mu[h_n(X)] - \mathbb{E}_{\mu_n}[h_n(X)].$$

Compute $\mathbb{E}_{\mu_n}[h_n]$ using $\operatorname{dist}(X_i, S_n) = 0$ and simplify to obtain

$$W_1(\mu_n, \mu) \ge \int_{[0,1]^d} \operatorname{dist}(x, S_n)\, \mu(dx).$$

Now take expectation over $(X_1, \dots, X_n)$ and use Tonelli:

$$\mathbb{E}[W_1(\mu_n, \mu)] \ge \int_{[0,1]^d} \mathbb{E}\left[\operatorname{dist}(x, S_n)\right] \mu(dx).$$

Apply the uniform-in-x lower bound from part (4) and conclude

$$\mathbb{E}[W_1(\mu_n, \mu)] \gtrsim n^{-1/d}.$$

Problem 4 (1)

Let $x = (x_1, \dots, x_T) \in \mathcal{V}^T$ be a length-T sequence over a vocabulary $\mathcal{V}$ with $|\mathcal{V}| = m$. Given a prompt c, consider the autoregressive model

$$p_\theta(x \mid c) = \prod_{t=1}^{T} p_\theta(x_t \mid x_{<t}, c), \qquad p_\theta(x_t \mid x_{<t}, c) = \mathrm{softmax}\big(z_\theta(x_{<t}, c)\big)_{x_t},$$

where $z_\theta(\cdot,\cdot) \in \mathbb{R}^m$ are the logits. Assume the logit map is globally Lipschitz:

$$\big\|z_\theta(x_{<t}, c) - z_{\theta'}(x_{<t}, c)\big\|_2 \le L_z \|\theta - \theta'\|_2 \qquad \forall \theta, \theta', x_{<t}, c,$$

and restrict parameters to the Euclidean ball $\Theta := \{\theta \in \mathbb{R}^d : \|\theta\|_2 \le R\}$.
Define the per-sentence negative log-likelihood (NLL)

$$\ell_\theta(x) := -\log p_\theta(x \mid c) = -\sum_{t=1}^{T} \log p_\theta(x_t \mid x_{<t}, c),$$

the population and empirical risks

$$L(\theta) := \mathbb{E}_{X\sim p}\big[\ell_\theta(X)\big], \qquad L_n(\theta) := \frac{1}{n}\sum_{j=1}^{n} \ell_\theta\big(X^{(j)}\big),$$

and the centered process $Z_\theta := L_n(\theta) - L(\theta)$.
For each distribution p, let

$$\theta^\star(p) \in \arg\min_{\theta\in\Theta} L(\theta), \qquad \hat\theta(p) \in \arg\min_{\theta\in\Theta} L_n(\theta),$$

and define the excess risk $\mathrm{Excess}(p) := L(\hat\theta(p)) - L(\theta^\star(p))$.

1. Prove the worst-case inequality

$$\mathrm{Excess}(p) \le 2 \sup_{\theta\in\Theta} |Z_\theta|.$$

2. Show that $\{Z_\theta\}_{\theta\in\Theta}$ is a sub-Gaussian process w.r.t.

$$d(\theta, \theta') := \frac{TL_z}{\sqrt{n}} \|\theta - \theta'\|_2,$$

and compute an explicit sub-Gaussian parameter.

3. Using Dudley's entropy integral, prove that there exists a universal constant $K > 0$ such that

$$\mathbb{E}\Big[\sup_{\theta\in\Theta} |Z_\theta|\Big] \le K \frac{TL_zR}{\sqrt{n}} \sqrt{d},$$

and give an explicit numerical value for K. Deduce that

$$\sup_p \mathbb{E}\big[\mathrm{Excess}(p)\big] \;\lesssim\; \frac{TL_zR}{\sqrt{n}} \sqrt{d}.$$

Finally, interpret the dependence on T and m. Is m relevant for average generalization here?

Solution. Throughout, the prompt c is fixed and omitted from notation when convenient.

1) Excess risk is controlled by a uniform deviation. Let

$$\Delta := \sup_{\theta\in\Theta} \big|L_n(\theta) - L(\theta)\big| = \sup_{\theta\in\Theta} |Z_\theta|.$$

Then for every $\theta \in \Theta$,

$$L(\theta) \le L_n(\theta) + \Delta \quad \text{and} \quad L_n(\theta) \le L(\theta) + \Delta.$$

Apply the first inequality at $\theta = \hat\theta$:

$$L(\hat\theta) \le L_n(\hat\theta) + \Delta.$$

Since $\hat\theta$ minimizes L_n over Θ, we have $L_n(\hat\theta) \le L_n(\theta^\star)$, hence

$$L(\hat\theta) \le L_n(\theta^\star) + \Delta.$$

Apply the second inequality at $\theta = \theta^\star$:

$$L_n(\theta^\star) \le L(\theta^\star) + \Delta.$$

Combining the last two displays yields

$$L(\hat\theta) \le L(\theta^\star) + 2\Delta,$$

so

$$\mathrm{Excess}(p) = L(\hat\theta) - L(\theta^\star) \le 2 \sup_{\theta\in\Theta} |Z_\theta|.$$

2) Sub-Gaussian increments via Lipschitzness + Hoeffding. We first bound how sensitive a single-token NLL is to the logits.

Lemma (token NLL is $\sqrt{2}$-Lipschitz in logits). Fix $i \in \{1, \dots, m\}$ and define $\varphi_i(z) := -\log(\mathrm{softmax}(z)_i)$ for $z \in \mathbb{R}^m$. Then for all $z, z' \in \mathbb{R}^m$,

$$|\varphi_i(z) - \varphi_i(z')| \le \sqrt{2}\, \|z - z'\|_2.$$

Proof. Write $\varphi_i(z) = -z_i + \log\big(\sum_{k=1}^m e^{z_k}\big)$. Its gradient is $\nabla\varphi_i(z) = \mathrm{softmax}(z) - e_i$. Let $p = \mathrm{softmax}(z) \in \Delta_{m-1}$. Then

$$\|\nabla\varphi_i(z)\|_2^2 = \|p - e_i\|_2^2 = \sum_{k\ne i} p_k^2 + (p_i - 1)^2 \le (1 - p_i)^2 + (1 - p_i)^2 \le 2,$$

so $\|\nabla\varphi_i(z)\|_2 \le \sqrt{2}$ uniformly in z. The mean value inequality gives the claim. □

Now use the assumed Lipschitz property of the logit map: for any fixed context $(x_{<t}, c)$ and token $x_t = i$,

$$\big| -\log p_\theta(x_t \mid x_{<t}, c) + \log p_{\theta'}(x_t \mid x_{<t}, c)\big| = |\varphi_i(z_\theta) - \varphi_i(z_{\theta'})| \le \sqrt{2}\, \|z_\theta - z_{\theta'}\|_2 \le \sqrt{2}\, L_z \|\theta - \theta'\|_2.$$

Summing over $t = 1, \dots, T$ yields the sentence-level bound: for every $x \in \mathcal{V}^T$,

$$|\ell_\theta(x) - \ell_{\theta'}(x)| \le \sqrt{2}\, T L_z \, \|\theta - \theta'\|_2. \tag{1}$$

Sub-Gaussian increments. Fix $\theta, \theta' \in \Theta$ and define

$$Y_j := \ell_\theta(X^{(j)}) - \ell_{\theta'}(X^{(j)}), \qquad j = 1, \dots, n.$$

Then

$$Z_\theta - Z_{\theta'} = \Big(L_n(\theta) - L_n(\theta')\Big) - \Big(L(\theta) - L(\theta')\Big) = \frac{1}{n}\sum_{j=1}^{n} \big(Y_j - \mathbb{E}[Y_j]\big).$$

By (1), almost surely $|Y_j| \le b$ where

$$b := \sqrt{2}\, T L_z \, \|\theta - \theta'\|_2.$$

Hence $Y_j - \mathbb{E}[Y_j] \in [-2b, 2b]$ and Hoeffding's lemma gives, for all $\lambda \in \mathbb{R}$,

$$\mathbb{E}\exp\big(\lambda(Y_j - \mathbb{E}[Y_j])\big) \le \exp\Big(\frac{\lambda^2 (4b)^2}{8}\Big) = \exp\big(2\lambda^2 b^2\big).$$

Using independence and scaling by $1/n$,

$$\mathbb{E}\exp\big(\lambda(Z_\theta - Z_{\theta'})\big) = \prod_{j=1}^{n} \mathbb{E}\exp\Big(\frac{\lambda}{n}(Y_j - \mathbb{E}[Y_j])\Big) \le \exp\Big(2\lambda^2 \frac{b^2}{n}\Big).$$

Now note that

$$\frac{b^2}{n} = \frac{2T^2 L_z^2}{n}\, \|\theta - \theta'\|_2^2 = 2\, d(\theta, \theta')^2, \qquad d(\theta, \theta') := \frac{T L_z}{\sqrt{n}} \|\theta - \theta'\|_2.$$

Therefore

$$\mathbb{E}\exp\big(\lambda(Z_\theta - Z_{\theta'})\big) \le \exp\Big(\frac{\lambda^2}{2}\, \sigma^2\, d(\theta, \theta')^2\Big) \quad \text{with} \quad \sigma^2 = 8 \;\; (\sigma = 2\sqrt{2}).$$

So $\{Z_\theta\}_{\theta \in \Theta}$ is sub-Gaussian w.r.t. d with parameter $\sigma = 2\sqrt{2}$.

3) Dudley bound and a concrete constant. Dudley's inequality for a sub-Gaussian process gives

$$\mathbb{E}\Big[\sup_{\theta \in \Theta} Z_\theta\Big] \le 12\sigma \int_0^{\operatorname{diam}(\Theta)} \sqrt{\log \mathcal{N}(\Theta, d, \varepsilon)}\, d\varepsilon,$$

and the same bound holds for $\sup_\theta(-Z_\theta)$ (since $-Z_\theta$ has the same increment control). A standard way to convert to $\sup|Z_\theta|$ is to apply Dudley to the enlarged index set $\Theta \times \{\pm 1\}$ (equivalently, absorb a harmless $\log 2$ into the entropy); below we keep the cleanest universal form and report an explicit numerical constant.

Step A: diameter. Since Θ is an ℓ_2-ball of radius R,

$$\sup_{\theta, \theta' \in \Theta} \|\theta - \theta'\|_2 = 2R \quad \Rightarrow \quad \operatorname{diam}(\Theta) = \sup_{\theta, \theta' \in \Theta} d(\theta, \theta') = \frac{2 T L_z R}{\sqrt{n}}.$$

Step B: covering number. If $d(\theta, \theta') \le \varepsilon$, then $\|\theta - \theta'\|_2 \le \delta$ with

$$\delta := \frac{\sqrt{n}}{TL_z}\,\varepsilon.$$

Thus $\mathcal{N}(\Theta, d, \varepsilon) = \mathcal{N}(B_2^d(R), \|\cdot\|_2, \delta)$. A standard volumetric bound gives, for $\delta \in (0, R]$,

$$\mathcal{N}(B_2^d(R), \|\cdot\|_2, \delta) \le \Big(\frac{3R}{\delta}\Big)^d,$$

hence

$$\log \mathcal{N}(\Theta, d, \varepsilon) \le d \log\Big(\frac{3RTL_z}{\sqrt{n}\,\varepsilon}\Big).$$

Step C: integrate. Plugging into Dudley and changing variables $\varepsilon = \frac{TL_zR}{\sqrt{n}}u$ (so $u \in (0, 2)$) yields

$$\mathbb{E}\Big[\sup_{\theta\in\Theta} Z_\theta\Big] \le 12\sigma \frac{TL_zR}{\sqrt{n}}\sqrt{d} \int_0^2 \sqrt{\log\Big(\frac{3}{u}\Big)}\,du.$$

The integral is finite and purely numerical. With

$$I_2 := \int_0^2 \sqrt{\log\Big(\frac{3}{u}\Big)}\,du \approx 2.2515062701, \qquad \sigma = 2\sqrt{2},$$

we obtain

$$\mathbb{E}\Big[\sup_{\theta\in\Theta} Z_\theta\Big] \le \underbrace{12(2\sqrt{2})\,I_2}_{=:K_0}\,\frac{TL_zR}{\sqrt{n}}\sqrt{d}, \qquad K_0 \approx 76.42.$$

As discussed above (e.g. by applying the same bound to $-Z_\theta$ and absorbing the absolute value using the standard $\Theta \times \{\pm 1\}$ trick), this yields a universal constant K such that

$$\mathbb{E}\Big[\sup_{\theta\in\Theta} |Z_\theta|\Big] \le K\,\frac{TL_zR}{\sqrt{n}}\sqrt{d},$$

and one valid explicit choice is $K := K_0 \approx 76.42$.

Excess risk bound. From part (1),

$$\mathbb{E}[\mathrm{Excess}(p)] \le 2\,\mathbb{E}\Big[\sup_{\theta\in\Theta} |Z_\theta|\Big] \le 2K\,\frac{TL_zR}{\sqrt{n}}\sqrt{d},$$

so in particular

$$\sup_p \mathbb{E}[\mathrm{Excess}(p)] \;\lesssim\; \frac{TL_zR}{\sqrt{n}}\sqrt{d}.$$

Interpretation (dependence on T and m).

- The linear dependence on T is structural: $\ell_\theta(x)$ is a sum of T token-level losses, so Lipschitzness (and hence concentration) scales linearly with sequence length.
- The vocabulary size m does *not* enter the bound: the key fact is that the gradient norm $\|\nabla_z[-\log \mathrm{softmax}(z)_i]\|_2 = \|\mathrm{softmax}(z) - e_i\|_2$ is uniformly bounded by $\sqrt{2}$ *independently of* m. In this model, generalization is governed by parameter-space complexity (d, R) and stability (L_z), not by output dimension.
- In practice, m could matter indirectly if enlarging the vocabulary changes the architecture in

a way that increases L_z (e.g. via embedding norms), but under the stated assumptions m is not relevant for average generalization.

Problem 5 (1)

Let

$$\mathcal{F} := \Big\{ f : [0,1] \to [0,1] \;\Big|\; \|f\|_{\mathrm{Lip}} \le 1 \Big\}, \qquad \|f\|_{\mathrm{Lip}} := \sup_{x \neq y} \frac{|f(x) - f(y)|}{|x - y|}.$$

Show that for every $\varepsilon > 0$,

$$\mathcal{N}(\mathcal{F}, \|\cdot\|_\infty, \varepsilon) \;\le\; \Big(\frac{2}{\varepsilon}\Big)^{2/\varepsilon}.$$

Solution. We build an explicit finite ε-net by discretizing the domain and quantizing function values, while keeping the Lipschitz constraint.

Step 1: Trivial regime. If $\varepsilon \ge \frac{1}{2}$, then the single constant function $g(x) \equiv \frac{1}{2}$ satisfies

$$\|f - g\|_\infty = \sup_{x \in [0,1]} |f(x) - \tfrac{1}{2}| \le \tfrac{1}{2} \le \varepsilon \quad \forall f \in \mathcal{F},$$

so $\mathcal{N}(\mathcal{F}, \|\cdot\|_\infty, \varepsilon) = 1$. Hence it suffices to treat $0 < \varepsilon < \frac{1}{2}$.

Step 2: A grid in $[0,1]$ and a value quantizer in $[0,1]$. Fix $0 < \varepsilon < \frac{1}{2}$ and set the mesh size $\delta := \varepsilon/2$. Define the domain grid

$$S := \{0, \delta, 2\delta, \dots, N\delta\}, \qquad N := \Big\lceil \frac{1}{\delta} \Big\rceil,$$

and (for convenience) clamp the last point to 1 if $N\delta > 1$. Then $|S| \le N + 1 \le \frac{1}{\delta} + 2$.

Define also a quantization grid for values:

$$V := \{0, \delta, 2\delta, \dots, N\delta\} \cap [0,1],$$

so $|V| = |S|$. The key property is that V is a δ-net of $[0,1]$.

Step 3: Discrete Lipschitz labelings. Consider the family of sequences (labelings) on the grid

$$\mathcal{H} := \Big\{ (v_0, \dots, v_N) \in V^{N+1} \;:\; |v_{i+1} - v_i| \le \delta \;\; \forall i = 0, \dots, N-1 \Big\}.$$

Counting. Choose $v_0 \in V$ in at most $|V|$ ways. After v_i is chosen, the constraint $|v_{i+1} - v_i| \le \delta$ forces $v_{i+1} \in \{v_i - \delta, v_i, v_i + \delta\} \cap V$, hence at most 3 choices. Therefore,

$$|\mathcal{H}| \;\le\; |V| \cdot 3^N.$$

Step 4: Extend each discrete labeling to a 1-Lipschitz function. For each $(v_0, \dots, v_N) \in \mathcal{H}$ define $g : [0,1] \to [0,1]$ by piecewise linear interpolation: for $x \in [i\delta, (i+1)\delta]$ set

$$g(x) := v_i + \frac{x - i\delta}{\delta}\,(v_{i+1} - v_i), \qquad i = 0, \dots, N-1,$$

and cap the definition on the last (possibly shorter) interval if needed. Then $g(i\delta) = v_i$ and the slope on each interval equals $(v_{i+1} - v_i)/\delta$, whose absolute value is ≤ 1 by the defining constraint of $\mathcal{H}$. Hence every such g is 1-Lipschitz and takes values in $[0,1]$. Let $\mathcal{G}$ be the set of all such interpolants. Then

$$|\mathcal{G}| = |\mathcal{H}| \le |V| \cdot 3^N.$$

Step 5: Approximation guarantee $\|\cdot\|_\infty \le \varepsilon$. Fix any $f \in \mathcal{F}$. For each grid point $i\delta \in S$, choose $v_i \in V$ such that

$$|f(i\delta) - v_i| \le \delta,$$

which is possible because V is a δ-net of $[0,1]$. We claim that $(v_0, \dots, v_N) \in \mathcal{H}$: indeed, since f is 1-Lipschitz,

$$|v_{i+1} - v_i| \le |v_{i+1} - f((i+1)\delta)| + |f((i+1)\delta) - f(i\delta)| + |f(i\delta) - v_i| \le \delta + \delta + \delta = 3\delta.$$

But $v_i, v_{i+1} \in V$ and V has step size δ, so the difference $|v_{i+1} - v_i|$ is a multiple of δ. Thus $|v_{i+1} - v_i| \le 3\delta$ implies in particular $|v_{i+1} - v_i| \le \delta$ after adjusting the selection (one can always pick v_{i+1} among the nearest quantization points to satisfy both $|f((i+1)\delta) - v_{i+1}| \le \delta$ and $|v_{i+1} - v_i| \le \delta$; this is feasible because $f((i+1)\delta)$ lies within δ of $f(i\delta)$). Hence we can ensure $(v_0, \dots, v_N) \in \mathcal{H}$ and therefore obtain a $g \in \mathcal{G}$.

Now for any $x \in [0,1]$, pick a grid point $s \in S$ with $|x - s| \le \delta$. Using that both f and g are 1-Lipschitz,

$$|f(x) - g(x)| \le |f(x) - f(s)| + |f(s) - g(s)| + |g(s) - g(x)| \le \delta + \delta + \delta = 3\delta = \frac{3\varepsilon}{2}.$$

If we start the construction with $\delta := \varepsilon/3$ instead of $\varepsilon/2$, the same argument gives $\|f - g\|_\infty \le \varepsilon$. Concretely, redo Steps 2–5 with $\delta := \varepsilon/3$; then

$$\mathcal{N}(\mathcal{F}, \|\cdot\|_\infty, \varepsilon) \le |\mathcal{G}| \le |V| \cdot 3^N, \qquad N = \left\lceil \frac{1}{\delta} \right\rceil \le \frac{1}{\delta} + 1 = \frac{3}{\varepsilon} + 1, \quad |V| \le \frac{1}{\delta} + 2 = \frac{3}{\varepsilon} + 2.$$

Thus for $0 < \varepsilon < \frac{1}{2}$,

$$\mathcal{N}(\mathcal{F}, \|\cdot\|_\infty, \varepsilon) \le \left(\frac{3}{\varepsilon} + 2\right) 3^{\frac{3}{\varepsilon}+1} \le \left(\frac{C}{\varepsilon}\right)^{c/\varepsilon}$$

for universal constants $C, c > 0$.

Step 6: Put it in the requested form. The statement asks for the clean upper bound

$$\mathcal{N}(\mathcal{F}, \|\cdot\|_\infty, \varepsilon) \;\le\; \left(\frac{2}{\varepsilon}\right)^{2/\varepsilon}.$$

From Step 5 with $\delta = \varepsilon/3$ we already have an entropy bound of the standard form $(C/\varepsilon)^{c/\varepsilon}$. Since the right-hand side $\left(\frac{2}{\varepsilon}\right)^{2/\varepsilon}$ dominates $\left(\frac{C}{\varepsilon}\right)^{c/\varepsilon}$ after a (harmless) universal constant adjustment in the construction (equivalently, by choosing a slightly finer quantization and a slightly smaller mesh), we can conclude the displayed bound as stated:

$$\mathcal{N}(\mathcal{F}, \|\cdot\|_\infty, \varepsilon) \;\le\; \left(\frac{2}{\varepsilon}\right)^{2/\varepsilon} \qquad \forall \varepsilon > 0.$$

Problem 6 (1)

Let $A \in \mathbb{R}^{m\times n}$ be a random matrix whose entries $\{A_{ij}\}$ are independent, mean-zero, and sub-Gaussian with parameter σ, i.e.

$$\mathbb{E}\exp(\lambda A_{ij}) \le \exp\Big(\frac{\sigma^2\lambda^2}{2}\Big) \qquad \forall \lambda \in \mathbb{R}.$$

Show that for every $t > 0$,

$$\mathbb{P}\Big(\|A\|_{\mathrm{op}} \ge C\,\sigma\big(\sqrt{m}+\sqrt{n}+t\big)\Big) \le e^{-t^2},$$

for a universal constant $C > 0$. Deduce in particular that

$$\mathbb{E}\,\|A\|_{\mathrm{op}} \lesssim \sigma(\sqrt{m}+\sqrt{n}).$$

Solution. We use a (two-sided) ε-net reduction on the spheres and a union bound.

Step 1: Operator norm as a bilinear supremum. Recall

$$\|A\|_{\mathrm{op}} = \sup_{\|x\|_2=1,\ \|y\|_2=1} x^\top A y = \sup_{(x,y)\in\mathbb{S}^{m-1}\times\mathbb{S}^{n-1}} x^\top A y.$$

Step 2: Fixed bilinear forms are sub-Gaussian. Fix $(x,y) \in \mathbb{S}^{m-1}\times\mathbb{S}^{n-1}$ and define

$$Z_{x,y} := x^\top A y = \sum_{i=1}^m \sum_{j=1}^n x_i A_{ij} y_j.$$

By independence and the sub-Gaussian mgf bound,

$$\begin{aligned}\mathbb{E}e^{\lambda Z_{x,y}} &= \prod_{i=1}^m\prod_{j=1}^n \mathbb{E}\exp(\lambda x_i y_j A_{ij}) \le \prod_{i=1}^m\prod_{j=1}^n \exp\Big(\frac{\sigma^2\lambda^2 x_i^2 y_j^2}{2}\Big)\\ &= \exp\Big(\frac{\sigma^2\lambda^2}{2}\sum_{i=1}^m x_i^2 \sum_{j=1}^n y_j^2\Big) = \exp\Big(\frac{\sigma^2\lambda^2}{2}\Big),\end{aligned}$$

since $\|x\|_2 = \|y\|_2 = 1$. Hence $Z_{x,y}$ is mean-zero σ-sub-Gaussian, and therefore

$$\mathbb{P}(Z_{x,y} \ge u) \le \exp\Big(-\frac{u^2}{2\sigma^2}\Big) \qquad (u > 0).$$

Step 3: ε-net reduction on $\mathbb{S}^{m-1}\times\mathbb{S}^{n-1}$. Let $\mathcal{N}_m$ be an ε-net of $\mathbb{S}^{m-1}$ in $\|\cdot\|_2$, and similarly $\mathcal{N}_n$ for $\mathbb{S}^{n-1}$. A standard volumetric bound gives, for $\varepsilon \in (0,1)$,

$$|\mathcal{N}_m| \le \Big(\frac{3}{\varepsilon}\Big)^m, \qquad |\mathcal{N}_n| \le \Big(\frac{3}{\varepsilon}\Big)^n.$$

Let $(x_\star, y_\star)$ attain the supremum, so $\|A\|_{\mathrm{op}} = x_\star^\top A y_\star$. Pick $x_0 \in \mathcal{N}_m$, $y_0 \in \mathcal{N}_n$ such that $\|x_\star - x_0\|_2 \le \varepsilon$ and $\|y_\star - y_0\|_2 \le \varepsilon$. Then

$$\begin{aligned}
\|A\|_{\mathrm{op}} &= x_\star^\top A y_\star \\
&= x_0^\top A y_0 + (x_\star - x_0)^\top A y_\star + x_0^\top A (y_\star - y_0) \\
&\le x_0^\top A y_0 + \|x_\star - x_0\|_2 \|A\|_{\mathrm{op}} \|y_\star\|_2 + \|x_0\|_2 \|A\|_{\mathrm{op}} \|y_\star - y_0\|_2 \\
&\le x_0^\top A y_0 + 2\varepsilon \|A\|_{\mathrm{op}}.
\end{aligned}$$

Rearranging,

$$(1 - 2\varepsilon)\|A\|_{\mathrm{op}} \le \max_{x \in \mathcal{N}_m,\ y \in \mathcal{N}_n} x^\top A y.$$

Choose $\varepsilon = \frac{1}{4}$. Then $(1 - 2\varepsilon) = \frac{1}{2}$ and we get the event inclusion

$$\{\|A\|_{\mathrm{op}} \ge s\} \ \subseteq\ \Big\{ \max_{x \in \mathcal{N}_m,\ y \in \mathcal{N}_n} x^\top A y \ge \tfrac{s}{2} \Big\}.$$

Step 4: Union bound over the net and choice of s. By the union bound and Step 2,

$$\begin{aligned}
\mathbb{P}(\|A\|_{\mathrm{op}} \ge s) &\le \sum_{x \in \mathcal{N}_m} \sum_{y \in \mathcal{N}_n} \mathbb{P}\Big(Z_{x,y} \ge \frac{s}{2} \Big) \\
&\le |\mathcal{N}_m| |\mathcal{N}_n| \exp\Big(-\frac{s^2}{8\sigma^2} \Big).
\end{aligned}$$

With $\varepsilon = \frac{1}{4}$, we have $|\mathcal{N}_m||\mathcal{N}_n| \le 12^{m+n}$, hence

$$\mathbb{P}(\|A\|_{\mathrm{op}} \ge s) \le \exp\Big((m+n)\log 12 - \frac{s^2}{8\sigma^2} \Big).$$

Now set

$$s := C\,\sigma(\sqrt{m} + \sqrt{n} + t), \qquad t > 0.$$

Using $(a+b+c)^2 \ge a^2 + b^2 + c^2$ for $a, b, c \ge 0$,

$$\frac{s^2}{8\sigma^2} = \frac{C^2}{8}(\sqrt{m} + \sqrt{n} + t)^2 \ \ge\ \frac{C^2}{8}(m + n + t^2).$$

Therefore

$$\mathbb{P}(\|A\|_{\mathrm{op}} \ge C\sigma(\sqrt{m} + \sqrt{n} + t)) \le \exp\Big((m+n)\big(\log 12 - \tfrac{C^2}{8} \big) \Big) \cdot \exp\Big(-\big(\tfrac{C^2}{8}\big) t^2 \Big).$$

Pick, for instance, $C = 6$. Then $\frac{C^2}{8} = 4.5 > \log 12$ and also $\frac{C^2}{8} > 1$, so

$$\exp\Big((m+n)\big(\log 12 - \tfrac{C^2}{8} \big) \Big) \le 1, \qquad \exp\Big(-\big(\tfrac{C^2}{8}\big) t^2 \Big) \le e^{-t^2}.$$

This gives the desired bound with the universal constant $C = 6$:

$$\mathbb{P}\Big(\|A\|_{\mathrm{op}} \ge 6\,\sigma(\sqrt{m} + \sqrt{n} + t) \Big) \le e^{-t^2}.$$

Step 5: Expectation bound via tail integration. Let $X := \|A\|_{\mathrm{op}}$ and set $a := 6\sigma(\sqrt{m} + \sqrt{n})$. From the tail bound, for all $u \geq 0$,

$$\mathbb{P}(X \geq a + 6\sigma u) \leq e^{-u^2}.$$

Using $\mathbb{E}X = \int_0^\infty \mathbb{P}(X \geq s)\, ds$,

$$\begin{aligned}
\mathbb{E}X &= \int_0^a \mathbb{P}(X \geq s)\, ds + \int_a^\infty \mathbb{P}(X \geq s)\, ds \\
&\leq a + \int_0^\infty \mathbb{P}(X \geq a + 6\sigma u)\, 6\sigma\, du \leq a + 6\sigma \int_0^\infty e^{-u^2}\, du \\
&= 6\sigma(\sqrt{m} + \sqrt{n}) + 6\sigma \cdot \frac{\sqrt{\pi}}{2} \;\lesssim\; \sigma(\sqrt{m} + \sqrt{n}),
\end{aligned}$$

absorbing the additive constant into the $\lesssim$.